

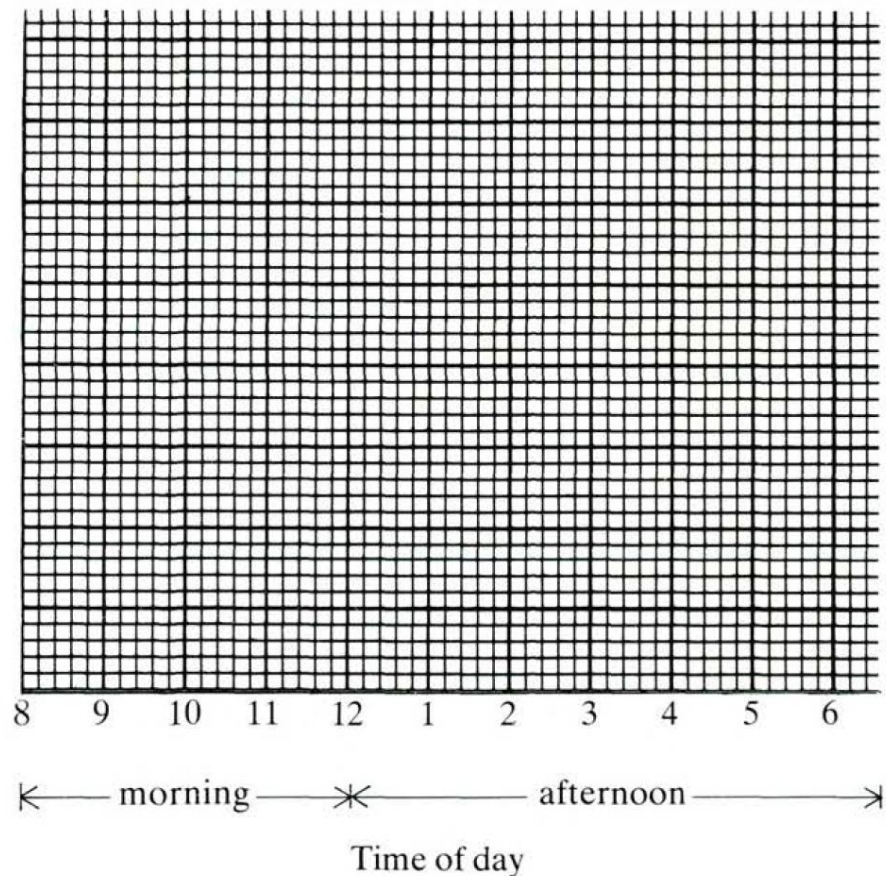
Worksheet: Graphs of physical processes

Part I: Cafeteria vending machine

A cafeteria has a vending machine selling sodas. On a typical day, the machine starts half full. No drinks are sold before 9AM and after 5PM. Sodas sell at a slow rate throughout the day except during the morning from 10:30AM to 11AM and at lunch break from 1PM to 2PM and the machine is filled once during the day right before the lunch break. It takes approximately 10 minutes to fill the machine.

1. Sketch a graph showing the amount of soda cans inside the vending machine at different times on a typical day below:

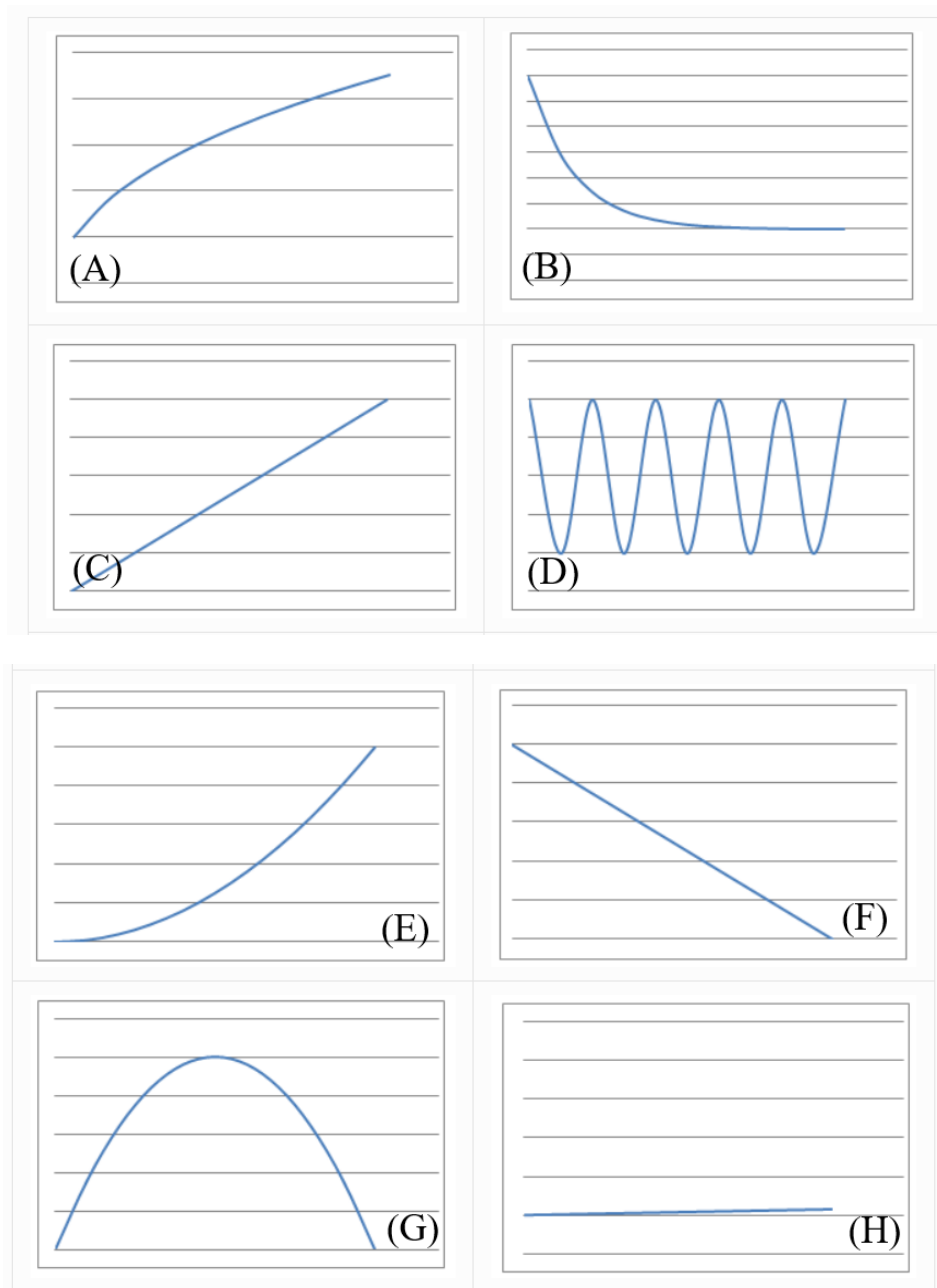
Number of
drinks
in the machine



2. Discuss with your peers and revise your graph
3. Visit https://xna24.github.io/math_teaching/grid_graph_drawing.html on your browser and use the web-based tool to sketch your revised graph and save to class shared Dropbox folder. Make sure to name your file using your first name + last initial, for instance "john_s.png".

Part II: Name that graph!

These are graphs of some physical processes

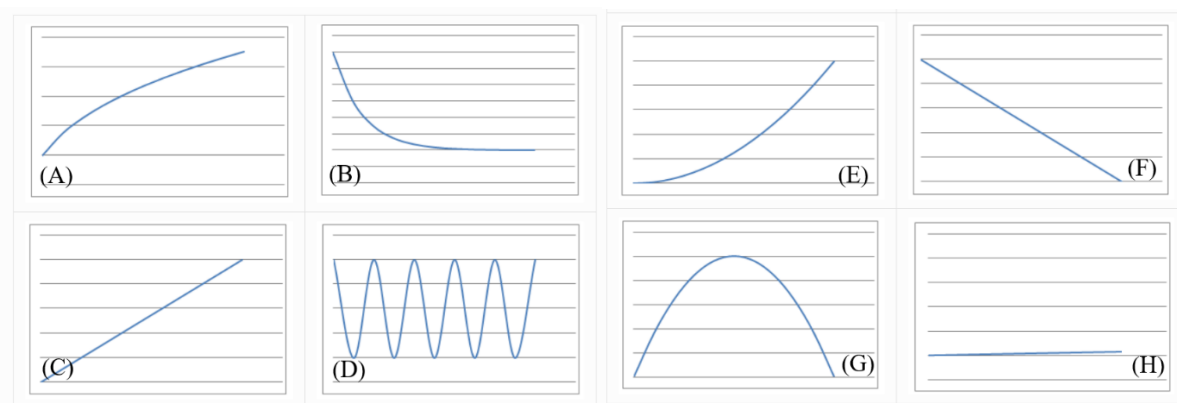


1. Can you suggest plausible processes which can give rise to each graph? Discuss with your peers and construct a list

2. The following is a list of 8 physical processes. Can you match them to the 8 given graphs?

- (1) A cup of tea is made and temperature measured in deg C every second. What would temperature-time graph look like?
- (2) The height of a valve on a bicycle tyre above the ground is measured after each centimetre that the bicycle travels forward. What would the height-distance graph look like?
- (3) I throw a tennis ball straight up into the air and catch it. The height of the ball from the ground is measured over the time of the journey using freeze-frame photography. What would height-time graph look like?
- (4) I measure several objects using inches and then using metres, plot them on a scatter graph, and join the points. What would the metres-inches graph look like?
- (5) I jump out of a plane and the distance fallen from the plain is measured every 0.1 sec until I open my parachute. What would distance fallen-time graph look like whilst in freefall?
- (6) I drive 70 miles an hour along a motorway and note the reading on my odometer (mile counter) every 5 minutes. What would the odometer reading-time graph look like?
- (7) I blow up a roughly spherical balloon using a balloon pump. After each pump, I measure the radius of the balloon in centimetres. What would the radius-number of pumps graph look like?
- (8) I suck water through a straw out of a large beaker at a constant rate and measure the volume of liquid remaining at various times. What would the volume-time graph look like?

3. Can you suggest equations that could model each process? Work individually first, then discuss with your peers before turning the page



4. Match each equation listed below to the graph and the related process. Justify your answer

$y = A\sqrt[3]{x}$	$y = A \sin(Bx) + A$	$y = Ax + B$	$y = \frac{A}{e^x} + B$
$y = Ax^2$	$y = Ax^2 + Bx + C$	$y = Ax$	$y = Ax + B$

5. Can you determine values for A, B, C for each equation? Note: for some graphs, you will need to make some assumptions.